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DEPARTMENT OF INFORMATICS

ICITDA 2022

ICITDA 2022

Proceedings of

The 2022 IEEE 7th International Conference on Information Technology and Digital Applications (ICITDA)

Yogyakarta, Indonesia
Hybrid Conference in November 4-5, 2022

Sponsored by:

IEEE INDONESIA SECTION

The Indonesia Section Computer Society Chapter

UNIVERSITAS ISLAM INDONESIA

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Welcome Message from Chair of ICITDA 2022

In the name of Allah, the Most Beneficent and the Most Merciful. May peace, mercy, and blessings of Allah be upon you.

Dear professors, colleagues, researchers, authors, and participants, please let me express my sincere gratitude and welcome you to the 2022 IEEE 7th International Conference on Information Technology and Digital Applications (ICITDA). Furthermore, on behalf of the committee members, I warmly welcome our keynote and workshop speakers, Prof. Dr. Ying-Dar Lin from National Yang-Ming Chiao-Tung University, Taiwan; Prof. Ts. Dr. Shukor Abd Razak from Universiti Sultan Zainal Abidin, Malaysia; Dr. Niken Dwi Wahyu Cahyani from Telkom University, Indonesia; and Dr. Ahmad Luthfi from Universitas Islam Indonesia, Indonesia. As you may already know, ICITDA is an annual international conference organized by the Department of Informatics, Faculty of Industrial Technology, Universitas Islam Indonesia, Indonesia. ICITDA 2022 is a hybrid (online-onsite) conference held on the 4th and 5th of November in Yogyakarta, Indonesia, with the theme “Bridging the Challenges in Network Security and Digital Forensics”.

Our hope and grand objective are to provide an effective forum for academicians, researchers, and practitioners to exchange ideas, knowledge, and experiences on the latest information technology and digital application development. This year we are delighted to have four keynote speakers and two workshop speakers from three countries. We also get valuable recommendations from reviewers from eight countries. Furthermore, there are over a hundred paper submissions from nine countries; forty-four papers from eight countries are accepted and presented.

ICITDA 2022 is the result of the hard work, support, and dedication of several parties. Therefore, I want to express my deep gratitude to all my teammates and the committee members who made the conference possible. The committee has worked diligently throughout the year to propose sessions, review a record number of submissions, answer inquiries, arrange the program schedule, and respond to last-minute requests. I also thank the Department of Informatics, Universitas Islam Indonesia, for all their support. Last but not least, I am very grateful to the paper submitters and reviewers who are the backbone of this conference.

Finally, thank you for being here and joining this conference. We are looking forward to seeing you next year at ICITDA 2023. Enjoy the conference.

Best regards,

Kurniawan D. Irianto
Chairman of the 2022 IEEE 7th ICITDA

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IEEE INDONESIA SECTION

UNIVERSITAS ISLAM INDONESIA

ICITDA 2022

PROGRAM BOOK ICITDA 2022

"Bridging The Challenges in Network Security and Digital Forensics"

November 4-5, 2022
Yogyakarta, Indonesia

Welcome Speech

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Chairman of the 2022 IEEE 7th ICITDA

PROGRAM SCHEDULE

November 4-5, 2022

IEEE INDONESIA SECTION

UNIVERSITAS ISLAM INDONESIA

ICITDA 2022

Program Schedule

Friday
November 4, 2022

Speaker



Prof. Ying-Dar Lin
National Yang-Ming Chiao-Tung University, Taiwan



Prof. Shukor Abd Razak
Universiti Sultan Zainal Abidin, Malaysia



Josua M. Sinambela, S.T., M.Eng.
CEO at RootBrain & ForensikDigital, Indonesia



Dr. Niken Dwi Wahyu Cahyani
Telkom University, Indonesia

Indonesia - Jakarta
Local Time(UTC+7)

Activities

07.00 - 08.00	Zoom opened
08.00 - 08.10	Opening Video
08.15 - 08.20	Opening speech by Committee chair
08.20 - 08.25	Opening speech by Chairman of IEEE Computer Society Indonesia Chapter
08.25 - 08.30	Opening speech by Head of Department of Informatics, Univ. Islam Indonesia
08.30 - 09.40	Keynote: **Prof. Dr. Ying-Dar Lin**, (National Yang-Ming Chiao-Tung University, Taiwan) Title: *AI for Cybersecurity: Attack Reproduction Multi-data source, Machine Learning, Data Balancing, MITRE ATT&CK Extension* Moderator: **Dr. Chandra Kusuma Dewa**, Universitas Islam Indonesia.
09.40 - 09.50	Coffee Break + UII Video Profile + Yogyakarta Video
09.50 - 11.00	Keynote: **Prof. Ts. Dr. Shukor Abd Razak** (Universiti Sultan Zainal Abidin, Malaysia) Title: *AR/VR Security and Privacy Risks* Moderator: **Dr. Syarif Hidayat**, Universitas Islam Indonesia.
11.00 - 13.00	Lunch Break
13.00 - 15.00	Workshop: **Josua M. Sinambela, S.T., M.Eng.** (CEO at RootBrain & Forensik Digital, Indonesia) Title: *Digital Forensic & Incident Response/Handling* Moderator: **TBA**
15.00 - 15.30	Short break
15.30 - 16.40	Keynote: **Dr. Niken Dwi Wahyu Cahyani** (Telkom University, Indonesia) Title: *Anti Forensics: A challenge in digital investigation* Moderator: **Mr. Ari Sujarwo,** Universitas Islam Indonesia

Program Schedule

Saturday
November 5, 2022

Speaker



Dr. Ahmad Luthfi
Universiti Islam Indonesia



Izazi Mubarok, S.S.T., M.Sc., CHFI, CEH, ACE, OFCE, MCFE, CCO, CCPA, CISA, CDSS, CRMP
Chairman of Indonesian Digital Forensic Association

Indonesia - Jakarta
Local Time(UTC+7)

Activities

08.00 - 10.00	Workshop: **Izazi Mubarok, S.S.T., M.Sc., CHFI, CEH, ACE, OFCE, MCFE, CCO, CCPA, CISA, CDSS, CRMP** Title: *Digital Forensic Investigation to Uncover Fraud Cases* Moderator: **Dr. Syarif Hidayat**, Universitas Islam Indonesia.
10.00 - 10.10	Coffee Break + UII Video Profile + Yogyakarta Video
10.10 - 11.55	Paper Presentation (Parallel Session)
11.55 - 13.00	Lunch Break
13.00 - 14.10	Keynote: **Dr. Ahmad Luthi** (Universitas Islam Indonesia) Title: *Validation of Digital Forensics Investigation: Challenges and Scientific Outlook* Moderator: **Mr. Ahmad Fathan Hidayatullah,** Universitas Islam Indonesia
14.10 - 14.20	Best Paper and Awards
14.20 - 14.30	Closing

PAPER PRESENTATION SCHEDULE

Parallel Session
November 5, 2022

IEEE INDONESIA SECTION
UNIVERSITAS ISLAM INDONESIA
ICITDA 2022

Room 2.12 (Captain Marvel) - Physical

Moderator: Ms. Erika Ramadhani

10.10 (UTC+7)
ID: 1570812610

A Bibliometrics Analysis of Digital Forensics Research in Indonesia Based on Scopus Index: 2012-2021
Erika Ramadhani; Dedy Hariyadi; Faulinda Ely Nastiti

10.25 (UTC+7)
ID: 1570824720

Implementation of Smart Contracts for E-Certificate as Non-Fungible Token Using Solana Network
Kevin Aditya Ramadhan Artha; Sarah Noor Zain; Azhar Ashhafa Alkautsar; Mochammad Haldi Widianto

10.40 (UTC+7)
ID: 1570825421

Development of Interactive Quiz Game Prototype as A Learning Evaluation Media
Harya Bima Dirgantara; Nur Choiriyati

10.55 (UTC+7)
ID: 1570826985

Performance of Intrusion Detection System Using Bagging Ensemble with SDN-Base Classifier
Amarudin Amarudin; Ridi Ferdiana; Widy Widyawan

Room 2.14 (Scarlet Witch) - Physical

Moderator: Ms. Lizda Iswari

10.10 (UTC+7)
ID: 1570823075

Evaluating the Village Website for Improving Its Quality
Agus Munandar

10.25 (UTC+7)
ID: 1570823962

Extracting Tourist Attraction Entities from Text Using Conditional Random Fields
Ahmad Fathan Hidayatullah; Rosyzie Apong; Daphne Teck Ching Lai; Atika Qazi

10.40 (UTC+7)
ID: 1570824475

SMOTE Classification and Random Oversampling Naive Bayes in Imbalanced Data
Nur Silviyah Rahmi

10.55 (UTC+7)
ID: 1570835874

Profiling the Spatial and Temporal Properties of Earthquake Occurrences Using ST-DBSCAN Algorithm
Lizda Iswari

Room 3.09 (Black Panther) - Virtual

Moderator: Ms. Novi Setiani

10.10 (UTC+7)
ID: 1570800268

Augmenting Light Supply in Urban Farming Using Artificial Lighting Based on Convolutional Neural Networks for Determination of Light Conditions of Lettuce

Stan K Holaysan; Irie Aldana; Christian Maderazo

10.25 (UTC+7)
ID: 1570814873

Maturity Grading and Macronutrient Deficiency Identification to Assist Tomato Growth in a Wick Hydroponic System Using an Xception-Based Convolutional Neural Network

Marc Andrei A. Buhawe; Arnold Loudel M. Rejuso; Christian V. Maderazo

10.40 (UTC+7)
ID: 1570824399

Software Testing Model by Measuring the Level of Accuracy Fault Output Using Neural Network Algorithm

Zulkifli Zlg; Ford Gaol; Agung Trisetyarso; Widodo Budiharto

10.55 (UTC+7)
ID: 1570825707

Data Collection with RPA to Build Virtual Assistant Laptop Recommendation with Recurrent Neural Network

Steven Tjayadi; Viny Christanti Mawardi

11.10 (UTC+7)
ID: 1570824156

Artificial Intelligence in Entrepreneurial Mindfulness Using CRISP-DM Method

Anthony Oktavius; Sasmoko Sasmoko; Yasinta Indrianti; Sonya Manalu; Jurike Moniaga

11.25 (UTC+7)
ID: 1570823276

Prediction of DHF (Dengue Hemorrhagic Fever) Severity Using Random Forest, kNN, Decision Tree and Naïve Bayes

Husni Iskandar Pohan; Yaya Heryadi; Lukas Lukas; Wayan Suparta; Antoni Wibowo

Room 4.11 (Captain America) - Virtual

Moderator: Mr. Hanson Prihantoro

10.10 (UTC+7)
ID: 1570824431

The Decision System for Selection of Zakat Mustahik Using Analytical Network Process Method

Dwi Asih Haryanti; Feni Andriani; Nurma Nugraha; Dewi Putrie Lestari; Beny Susanti

10.25 (UTC+7)
ID: 1570825892

Classification of Flood Alert in Jakarta with Random Forest

Febryan Grady; Joelry Tarigan; Joel Wahidiyat; Anang Prasetyo

10.40 (UTC+7)
ID: 1570821405

Understanding Social Media Behavior in Philippines Presidential Election Using Natural Language Processing

Ken Gorro; Leodivino Lawas; Rosein Ancheta; Anthony Ilano

10.55 (UTC+7)
ID: 1570812414

Malay Sarcasm Detection on Social Media: A Review, Taxonomy, and Future Directions

Suziane Haslinda Suhaimi; Nur Azaliah Abu Bakar; Nurulhuda Firdaus Mohd Azmi

11.10 (UTC+7)
ID: 1570814953

Systematic Literature Review of Stemming and Lemmatization Performance for Sentence Similarity

Rio Pramana; Debora Debora; Jonathan J Subroto; Alexander Agung Santoso Gunawan; Anderies Anderies

11.25 (UTC+7)
ID: 1570821348

Building Damage Assessment Using Deep Learning: Bibliometric Analysis

Sidharta Sidharta

11.40 (UTC+7)
ID: 1570805020

Hypersonic Flight Control Using Vector Field Method - Straight Line Path

Muhammad Ridho Rosa; Muhammad Faizi; Erwin Susanto

Room 3.09 (Black Panther) - Virtual

Moderator: Mr. Kholid Haryono

10.10 (UTC+7)
ID: 1570816216

Web Page Blocker Using Machine Learning Classifier and Chrome Plugin

Ken Gorro; Larmie Feliscuzo; Cherry Lyn Sta Romana

10.25 (UTC+7)
ID: 1570815166

Type of Training Recommendation Based on Body Fat Prediction Using LASSO Regression

Teddy Lioner; Dyah Herwindiati; Janson Hendryli

10.40 (UTC+7)
ID: 1570816250

An Identification and Analysis of Fake News About COVID-19 Vaccines in the Philippines by Using a Classification Model

Margaret Alexis L. Oquias; Angie Ceniza-Canillo; Riane Nichelle Daeve I. Acharon

10.55 (UTC+7)
ID: 1570824081

Systematic Literature Review: Toxic Comment Classification

Felix Museng; Adelia Jessica; Nicole Wijaya; Anderies Anderies; Irene Anindaputri Iswanto

11.10 (UTC+7)
ID: 1570815704

Sentiment Analysis of Hog Raisers During African Swine Fever Using Vader Lexicon-Based Methods

Evelyn Baesa; Edralin R. Raro; Victor Q. Parillas, Jr.; Jocelyn M Beriña; Thelma Palaoag

11.25 (UTC+7)
ID: 1570818251

The Design Science Research Methodology for Self Assessing Digital Transformation Maturity Index in Developing Countries

Tining Haryanti; Nur Aini Rakhmawati; Apol Pribadi Subriadi; Aris Tjahyanto

Room 3.11 (Spider Man) - Virtual

Moderator: Mr. Rian Adam R

10.10 (UTC+7)
ID: 1570816606

Sentiment Analysis of Product Reviews as Customer Recommendations in Shopee Philippines Using Hybrid Approach

Vin Myca C. Sagarino; Jennen Isabelle M. Montejo; Angie Ceniza-Canillo

10.25 (UTC+7)
ID: 1570821565

Sentiment Analysis of Nationwide Lockdown amid COVID 19: Evidence from Pakistan

Atika Qazi; Javaria Qazi; Khulla Naseer; Muhammad Zeeshan; Baseer Khan; Samrat Kumar Dey

10.40 (UTC+7)
ID: 1570813869

Discovering Philippine's Occupational Gender Segregation Through Data Analytics

Christian Y Sy; Mary Joy P Canon; Thelma Palaoag; Lany Maceda; Rachel Edita Roxas

10.55 (UTC+7)
ID: 1570834977

What Can We Learn from MOOC: A Sentiment Analysis, n-Gram, and Topic Modeling Approach

Timothy Dillan; Dhomas Hatta Fudholi

11.10 (UTC+7)
ID: 1570835061

Rich Kid, Poor Kid: Students Clustering Using K-Prototype Algorithm

Sabar Aritonang Rajagukguk; Dhomas Hatta Fudholi

11.25 (UTC+7)
ID: 1570821823

Classifying COVID-19 Lung CT-Scans Using Complex-Valued CNN

Yingqi Pei

11.40 (UTC+7)
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KEYNOTE SPEAKER

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Keynote 1

Prof. Ying-Dar Lin

National Yang-Ming Chiao-Tung University, Taiwan

AI for Cybersecurity: Attack Reproduction, Multi-datasource Machine Learning, and MITRE ATT&CK Extensions

Intrusions have increasingly long kill chains, with various footprints left for detection and investigation, and thus require machine learning to better recognize them. These footprints include threat intelligence in IP's and domains, network sighting in packets, and host sighting in logs and others. Relying on one single data source might result in more false negatives. In this talk, we first argue why AI for cybersecurity and why multiple data sources. Then we introduce a toolchain, CRÈME, we developed to complete the cycle of attack reproduction, multi-dataset collection, machine learning, and F1-score evaluation. With experimental results, we show how and why traffic, log, and statistics combined together could enhance F1 score. The tool serves as a platform for further research. We highlight several important steps towards the MITRE ATT&CK framework where both attack techniques and kill chains (lifecycles) should be recognized.



Keynote 2

Prof. Shukor Abd Razak

Universiti Sultan Zainal Abidin, Malaysia

AR Security and Privacy Risk

In AR, users experience realism of virtual and augmented reality by utilizing their biometric data such as iris or retina scans, fingerprints, face geometry, handprints and voiceprints. Users need to keep in mind that there are still many loopholes in AR technologies that could be exploited by hackers to collect personal data. For instance AR technologies can see everything what the user sees and this for sure impose security risks to the users. The talk will discuss security and privacy issues related to the usage of AR in various perspectives and provide suggestions to minimize the risks.



Keynote 3

Dr. Niken Dwi Wahyu Cahyani

Telkom University, Indonesia

Anti Forensics: A Challenge in Digital Investigation

Digital forensics is very essential in any digital or physical investigation where data is involved after an incident. This data, whether it is personal or organizational, public or confidential, plays an important role in uncovering cases. The aim of digital forensics is to legally acquire and analyze the data, while anti-forensics aims to hinder forensics by complicating its activities. In this presentation, we will identify the coverage of digital forensics, their challenges in technical and non-technical aspects including operational, legal, and investigative challenges, and also explore current anti-forensics techniques and the available countermeasures.



Keynote 4

Dr. Ahmad Lutfi
Universitas Islam Indonesia

Validation of Digital Forensics Investigation: Challenges and Scientific Outlook

One of the most essential aspects of digital forensics is ensuring the integrity of the data collected so that they can be presented as valid evidence in court. The presentation for ICITDA 2022 is based on several occurrences. First off, the practitioner-driven or professional sector continues to play a dominant role in the development and study of the area of digital forensics. Second, studies in the field of digital forensics tend to be ad hoc, investigator-driven, and based on individual experience. Given this context, the digital forensic analysis still has a large number of flaws and vulnerabilities, particularly during the validation stage. Therefore, taking into account the challenges and scientific outlook, this study proposes a validation model with a tested and systematic approach.



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Assessment of a Learning Management System as a Blended Workplace Using the UTAUT2 Model

Kennedy Centeno Cuya; Thelma Palaoag

As we transitioned to the new normal, most academic institutions implemented a comprehensive work-from-home strategy during the height of the pandemic. An established research model will be used to evaluate an LMS as a blended workplace, with the goal of establishing a reference point. Likert scale with indication of 1 as strongly disagree to 7 as strongly agree, the questionnaire survey was designed in accordance with the established research model of UTAUT2 by Venkatesh. Structural Equation Modeling and Partial Least Squares (PLS) were used to analyze the data in this study. This study assessed the reliability and validity of the proposed hypotheses using structural model, measurement model, and moderator analysis. Other confirmatory analyses were performed to assess the developed hypothesis, including goodness of fit, goodness of model, collinearity, bootstrapping, and its predictive power. It was found out that the factors included in the proposed research framework of the study had a positive and significant impact on the use of the LMS as a blended workplace application in the study. As a result, it is recommended that it be used for university-wide implementation. A learning management system (LMS), specifically Moodle, can be extremely beneficial for educational institutions that are applying for international standard certification or program accreditation. It can assist with maintaining administrative tasks, submission, reviewing, approval, and monitoring of required documents.

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Augmented Reality in Chemistry Learning: A Systematic Literature Review

Hani Dewi Ariessanti

This study reviews factors affecting AR-based chemistry learning with the BIM model. Although there have been many kinds of research explaining the application of augmented reality (AR) and building information modeling (BIM) in learning, no research has been found that discusses the factors affecting the integration of BIM in AR-based learning. This study adopted the SLR method with the PRISMA protocol by scanning articles written in English or articles written in Indonesian from accredited online databases, and fully accessible by the authors. The articles reviewed were related to AR, BIM, Virtual Reality (VR), and chemistry learning. The predefined strings were used to extract the 156 articles. Only 41 articles met the inclusion criteria that were used as the reference for this study. Six factors were found that affect the AR-based chemistry learning with the BIM model. They are AR work performance, accessibility, user interaction with the system, presentation of material, understanding, and models. This study implies that the factors extracted can be considered as a model in developing learning media to improve chemistry learning. The limitation of this study is the absence of further analysis of which critical factors should be addressed. Researchers recommend further research by reviewing more literature.



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